

Gebze Technical University

Dep. of Mechanical Engineering

Statics (ME 231)

Lecture 14:

Friction in Applications

By Saeed Lotfan

Friction in Applications: Wedges

A **wedge** is a simple machine that is often used to transform an applied force into much larger forces. Wedges can also be used to slightly move or adjust heavy loads.

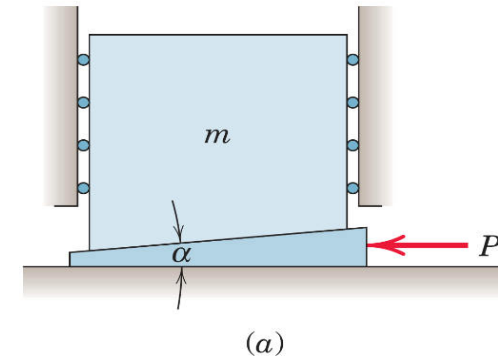


Wedges are often used to adjust the elevation of structural or mechanical parts. Also, they provide stability for objects such as this pipe.

Friction in Applications: Wedges

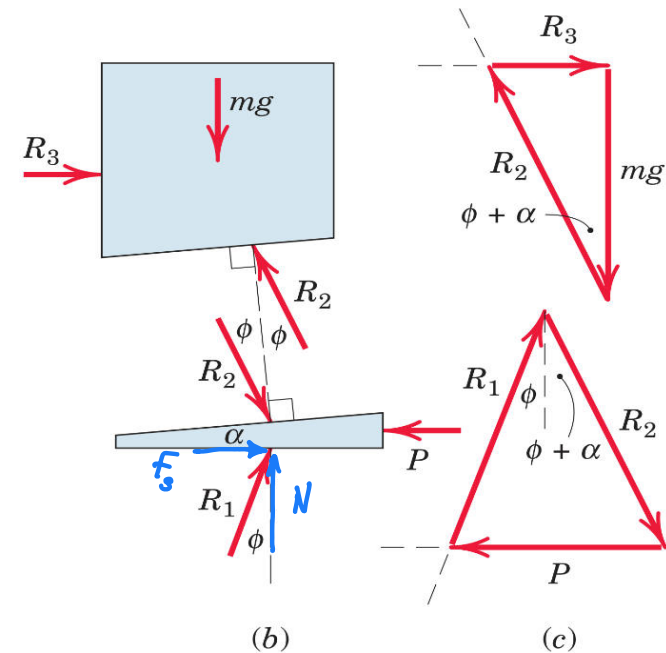
Free-Body diagram ($\alpha \leq \phi$):

If P is removed and the wedge remains in place, equilibrium of the wedge requires that the equal reactions R_1 and R_2 be collinear.



$$\tan \phi = \frac{F_s}{N}$$

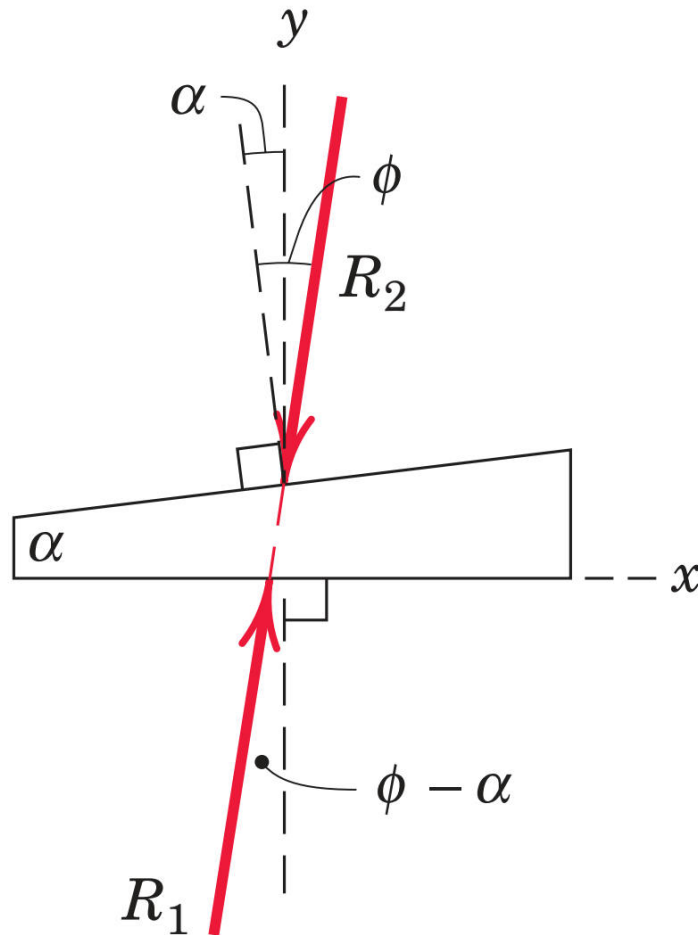
$$\tan \phi_s = \frac{\mu_s N}{N} = \mu$$



Forces to raise load

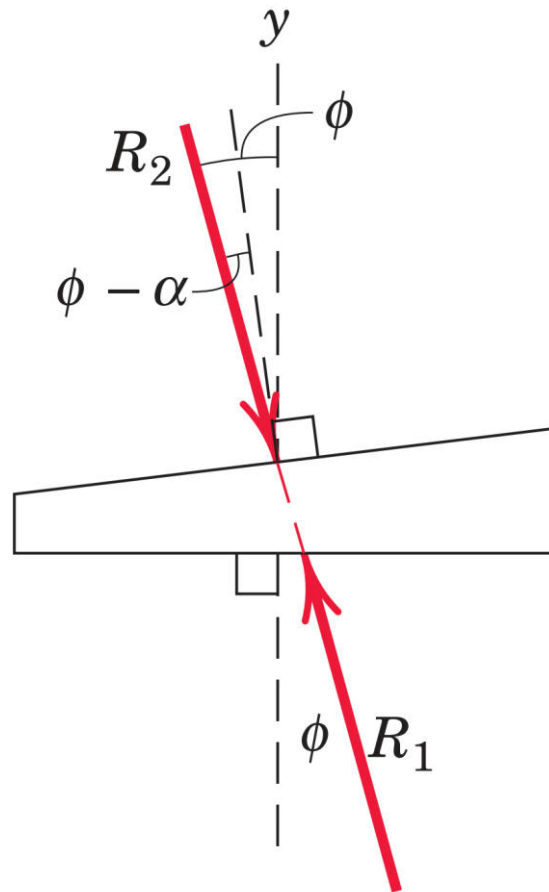
Friction in Applications: Wedges

Impending slippage at the upper surface of Wedge ($\alpha \leq \phi$):



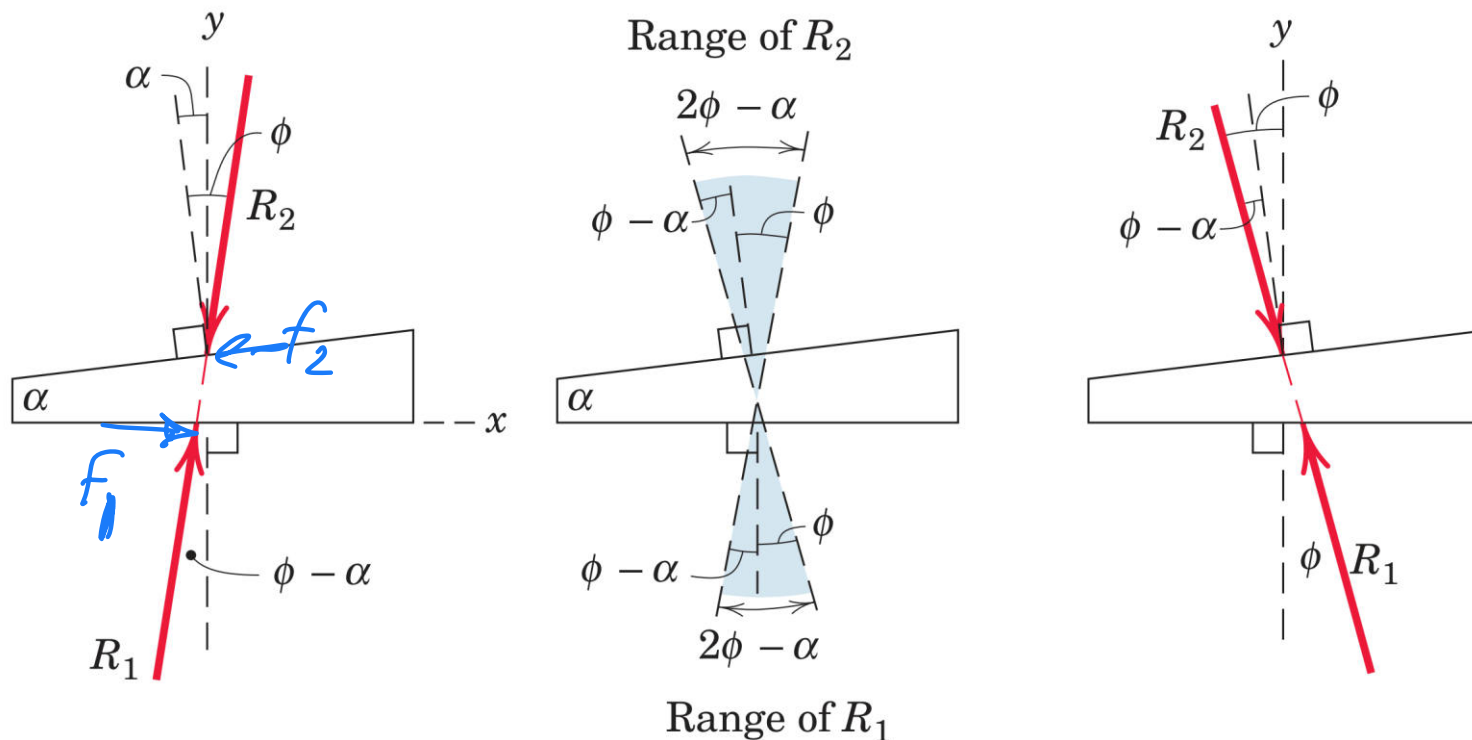
Friction in Applications: Wedges

Impending slippage at the lower surface of Wedge ($\alpha \leq \phi$):



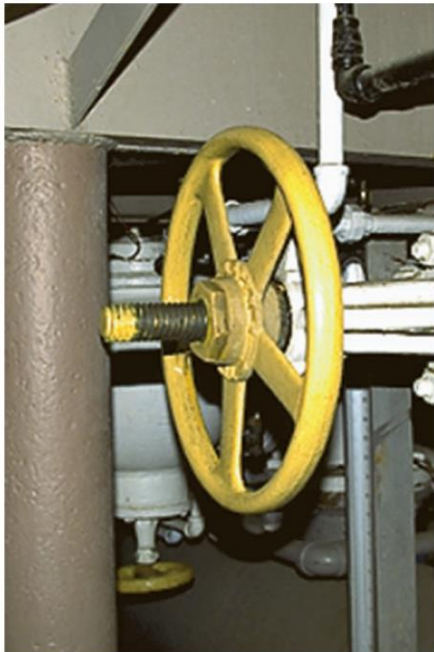
Friction in Applications: Wedges

For the wedge to slide out of its space, slippage must occur at **both** surfaces simultaneously; otherwise, the wedge is **self-locking**, and there is a finite range of possible intermediate angular positions of R_1 and R_2 for which the wedge will remain in place. **That is when $\alpha < 2\phi$.**



Friction in Applications: Screws

In most cases, screws are used as fasteners; however, in many types of machines they are incorporated to transmit power or motion from one part of the machine to another. A **square-threaded screw** is commonly used for this purpose, especially when large forces are applied along its axis.



Square-threaded screws find applications on valves, jacks, and vises, where particularly large forces must be developed along the axis of the screw.

Friction in Applications: Screws

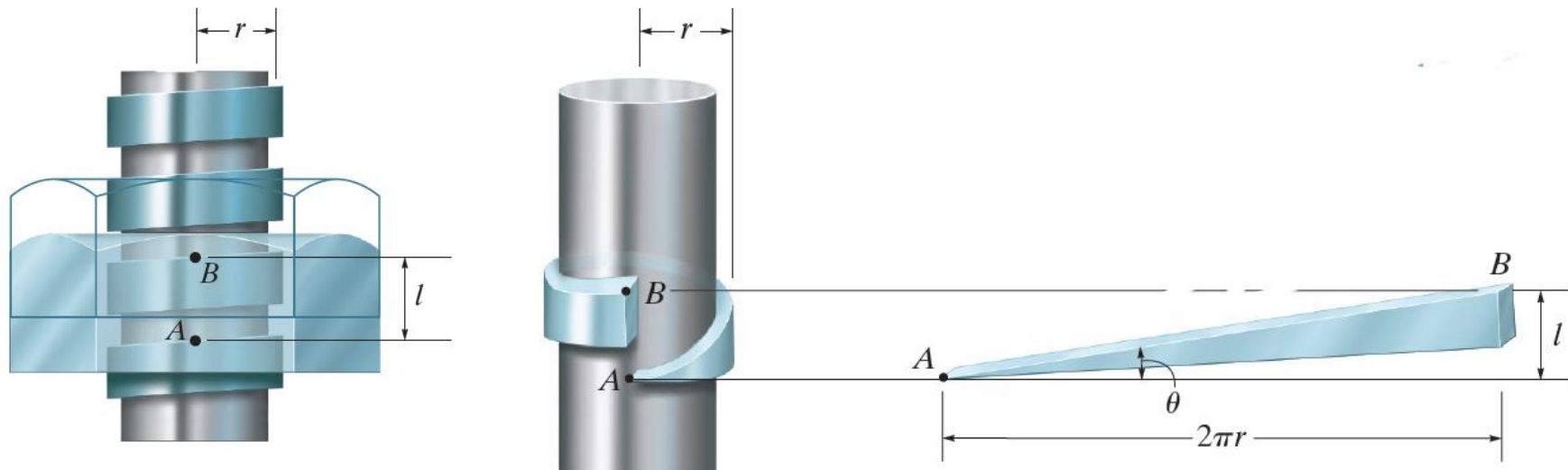
Geometric Properties:

l is the **lead** of the screw.

r is the **mean radius** of the thread.

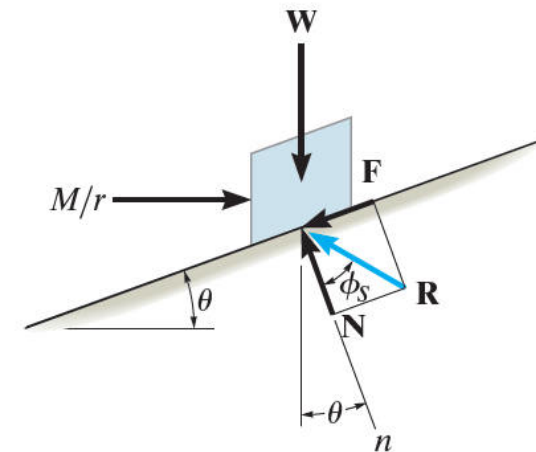
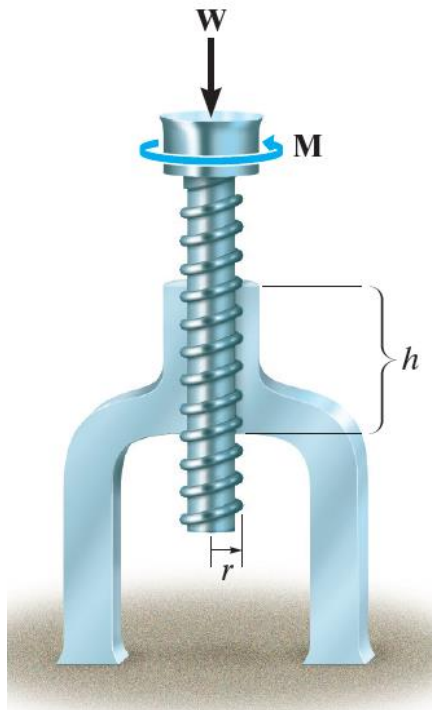
θ is the **lead angle** of the screw.

$$\theta = \tan^{-1}(l/2\pi r)$$



Friction in Applications: Screws

Upward Impending Motion (Against W)



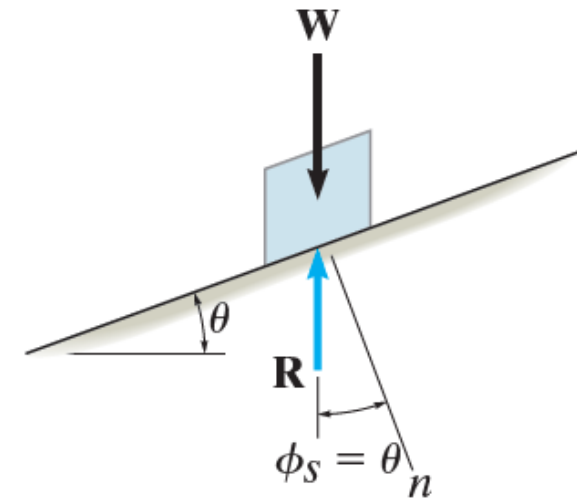
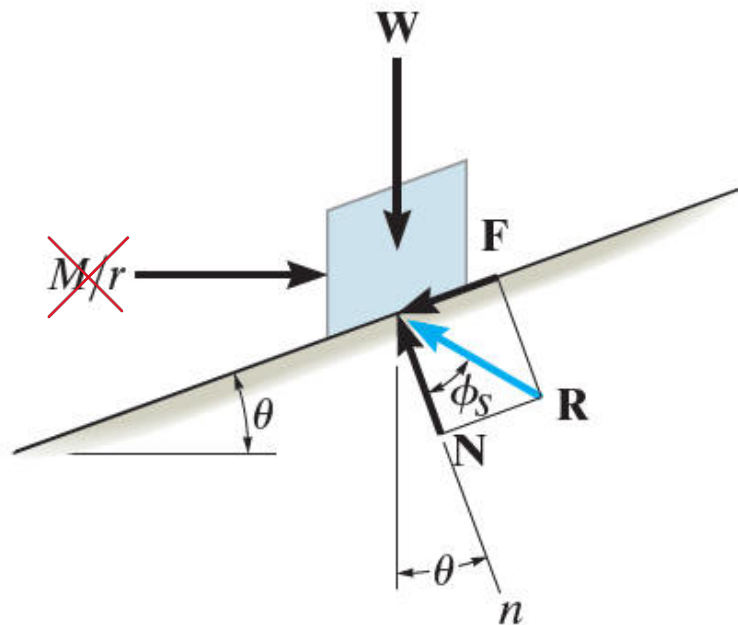
Upward screw motion

$$\begin{aligned} \rightarrow \Sigma F_x &= 0; & M/r - R \sin (\theta + \phi_s) &= 0 \\ + \uparrow \Sigma F_y &= 0; & R \cos (\theta + \phi_s) - W &= 0 \end{aligned}$$

$$M = rW \tan (\theta + \phi_s)$$

Friction in Applications: Screws

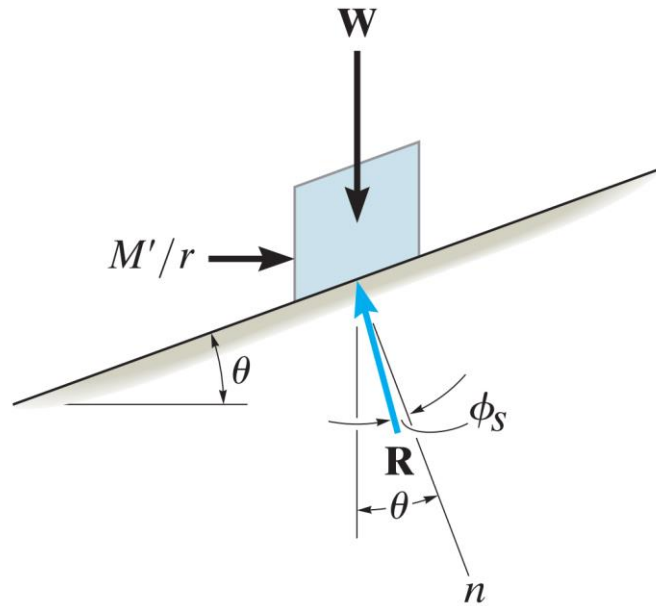
Self-Locking Screw



Self-locking screw ($\theta = \phi_s$)
(on the verge of rotating downward)

Friction in Applications: Screws

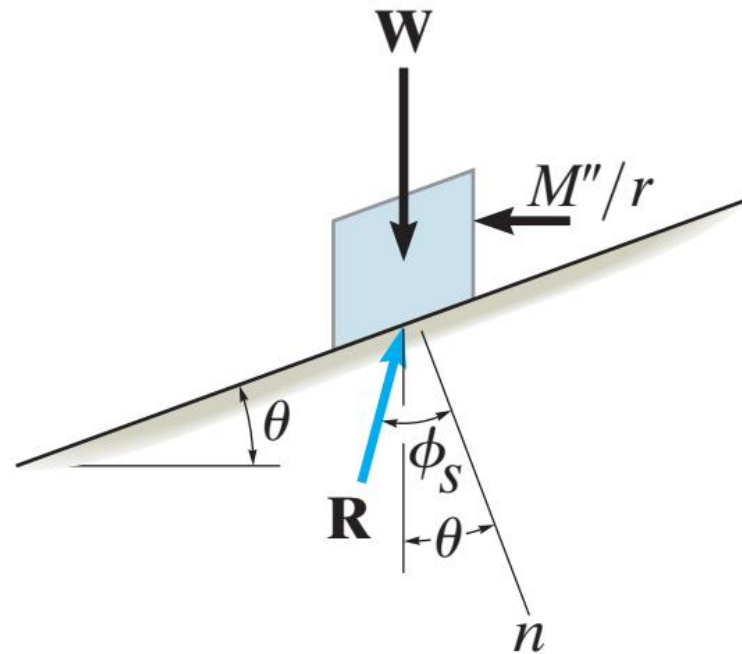
Downward Impending Motion, ($\theta > \phi_s$)



$$M' = rW \tan (\theta - \phi_s)$$

Friction in Applications: Screws

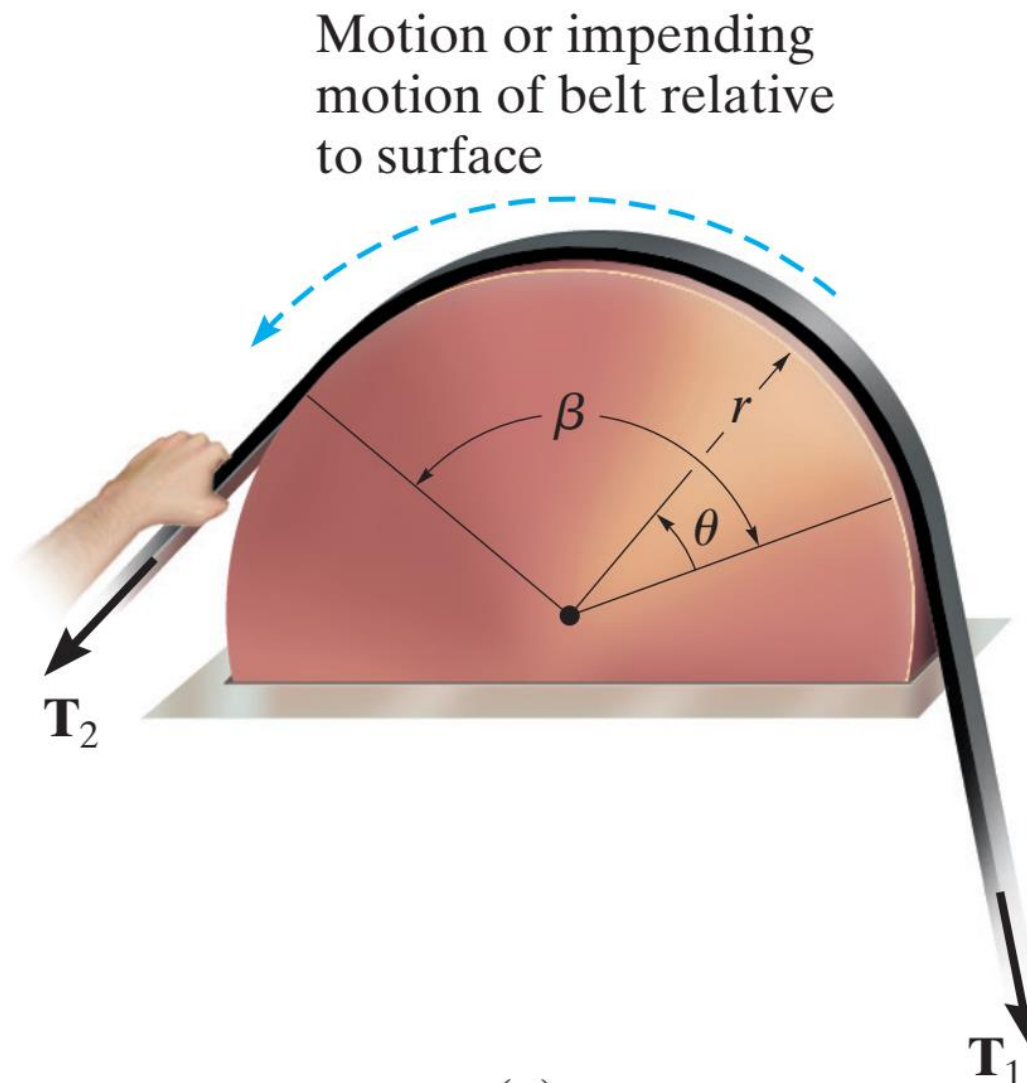
Downward Impending Motion, ($\theta < \phi_s$)



$$M'' = rW \tan (\phi_s - \theta)$$

This moment is necessary to overcome the friction and self-locking!

Friction in Applications: Flat Belts



Friction in Applications: Flat Belts

Frictional analysis

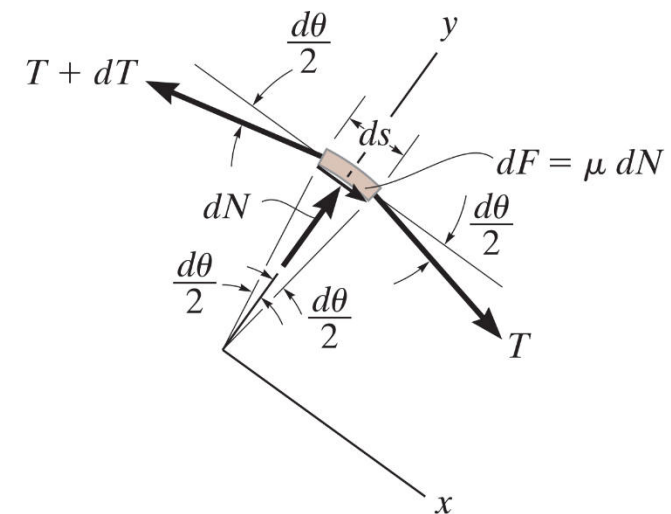
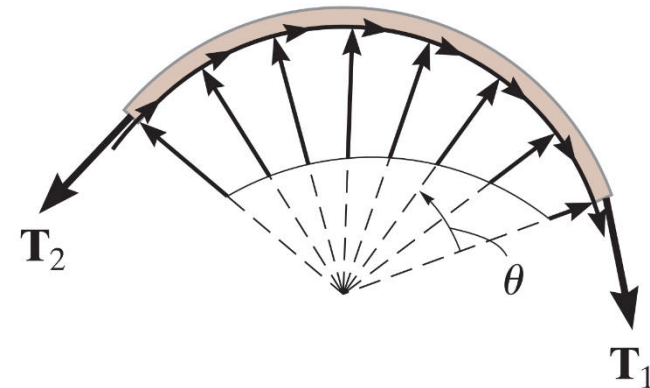
$$\begin{aligned} \sum F_x = 0; & \quad T \cos\left(\frac{d\theta}{2}\right) + \mu dN - (T + dT) \cos\left(\frac{d\theta}{2}\right) = 0 \\ \sum F_y = 0; & \quad dN - (T + dT) \sin\left(\frac{d\theta}{2}\right) - T \sin\left(\frac{d\theta}{2}\right) = 0 \end{aligned}$$

$\mu dN = dT$
 $dN = T d\theta$

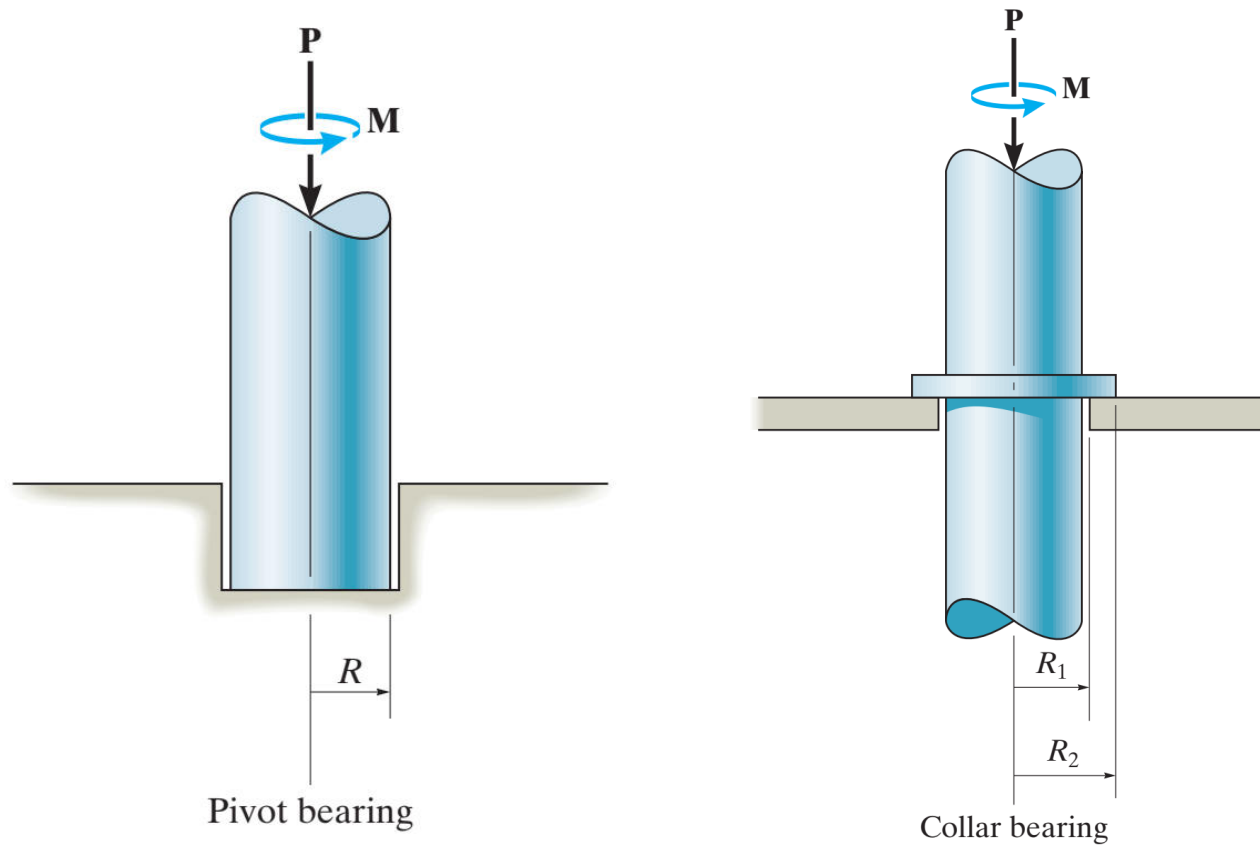
Then:

$$T_2 = T_1 e^{\mu\theta}$$

$\mu T d\theta = dT \rightarrow \int \frac{dT}{T} = \int \mu d\theta$
 $\ln T \Big|_{T_1}^{T_2} = \mu\theta \rightarrow T_2 = T_1 e^{\mu\theta}$



Friction in Applications: Collar Bearings, Pivot Bearings, and Disks (Thrust support)

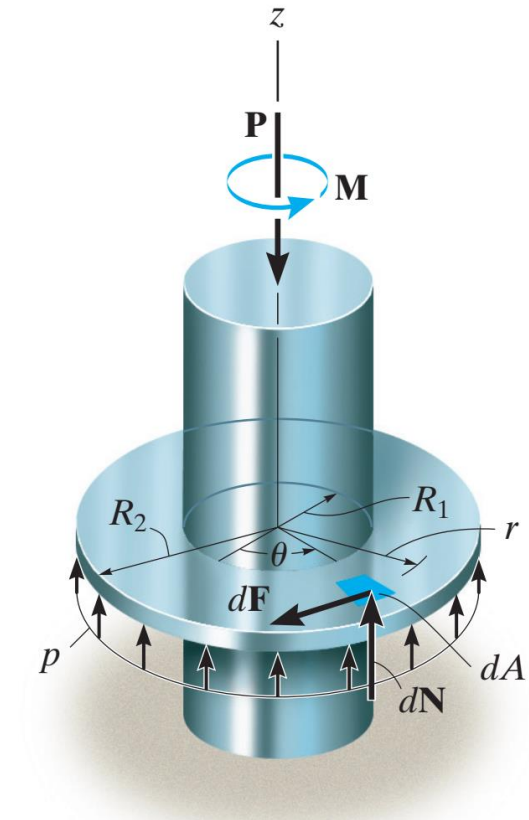


Friction in Applications: Collar Bearings, Pivot Bearings, and Disks (Thrust support)

Frictional Analysis

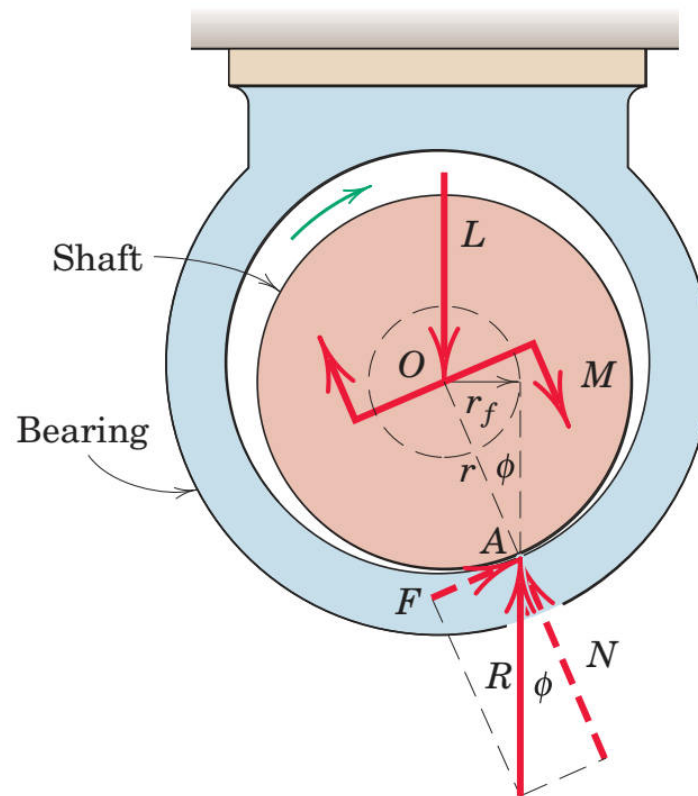
$$dF = \mu_s dN = \mu_s p dA = \frac{\mu_s P}{\pi(R_2^2 - R_1^2)} dA$$

$$\Sigma M_z = 0; \quad M - \int_A r dF = 0$$



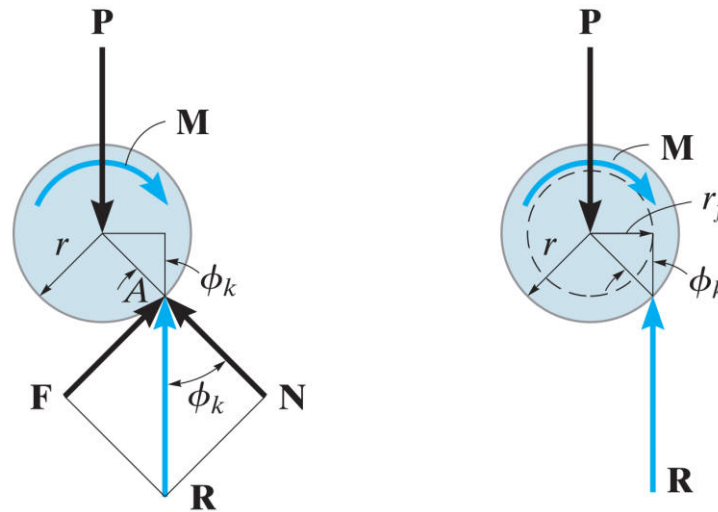
Friction in Applications: Journal Bearings

A **journal bearing** is one which gives lateral support to a shaft in contrast to axial or thrust support. For dry bearings and for many partially lubricated bearings we may apply the principles of dry friction.



Friction in Applications: Journal Bearings

Frictional Analysis



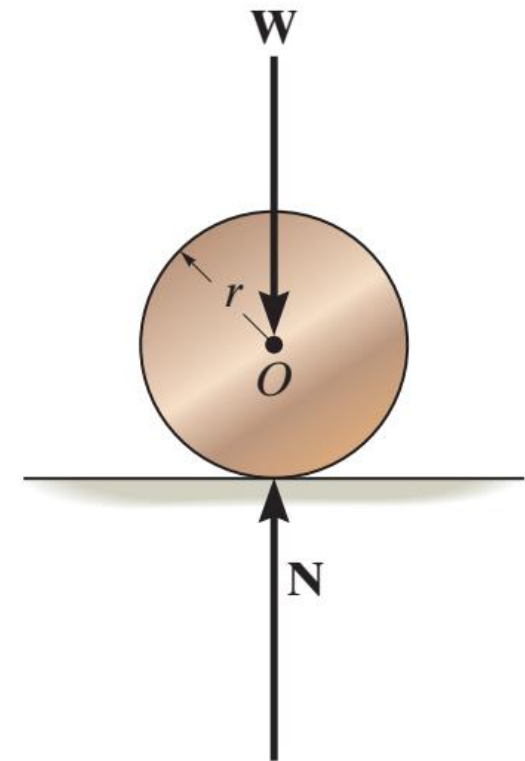
$$M = Rr \sin \phi_k$$

Here, ϕ_k is the angle of kinetic friction and $r \sin \phi_k = r_f$. The dashed circle with this radius is called the **friction circle**, and as the shaft rotates, the reaction R will always be tangent to it.

Rolling Resistance

Idealization

When a *rigid* cylinder rolls at constant velocity along a *rigid* surface, the normal force exerted by the surface on the cylinder acts perpendicular to the tangent at the point of contact.

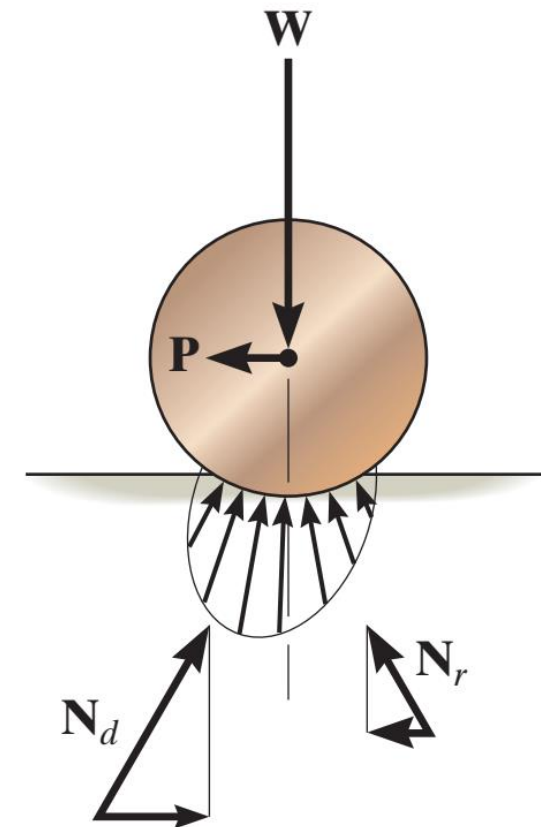


Rigid surface of contact

Rolling Resistance

Now, consider the cylinder to be made of a very hard material, and the surface upon which it rolls to be relatively soft. As the cylinder of weight $\mathbf{W}$ rolls, the surface material in front of the cylinder *retards* (delays) the motion since it is being *deformed*, whereas the material in the rear is *restored* from the deformed state, and therefore tends to *push* the cylinder forward.

The magnitude of the force of *deformation*, $\mathbf{N}_d$, and its horizontal component is *always greater* than that of *restoration*, $\mathbf{N}_r$, and consequently a horizontal driving force $\mathbf{P}$ must be applied to the cylinder to maintain the motion.

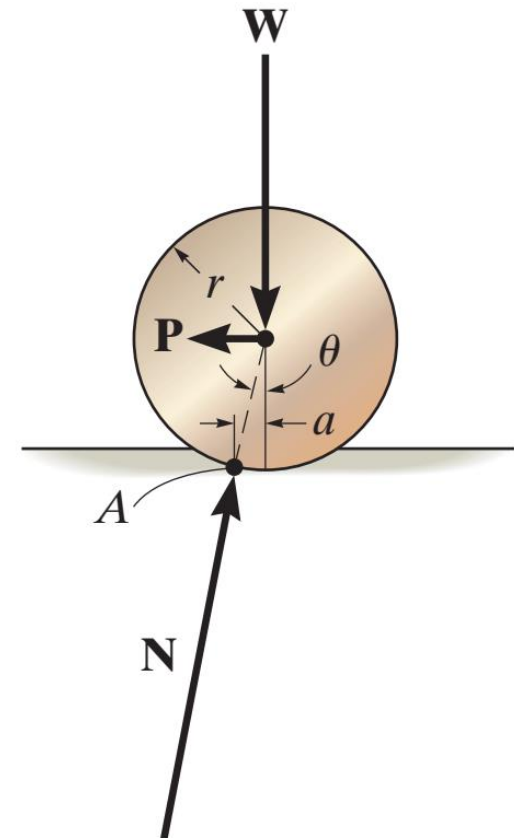


Soft surface of contact

Rolling Resistance

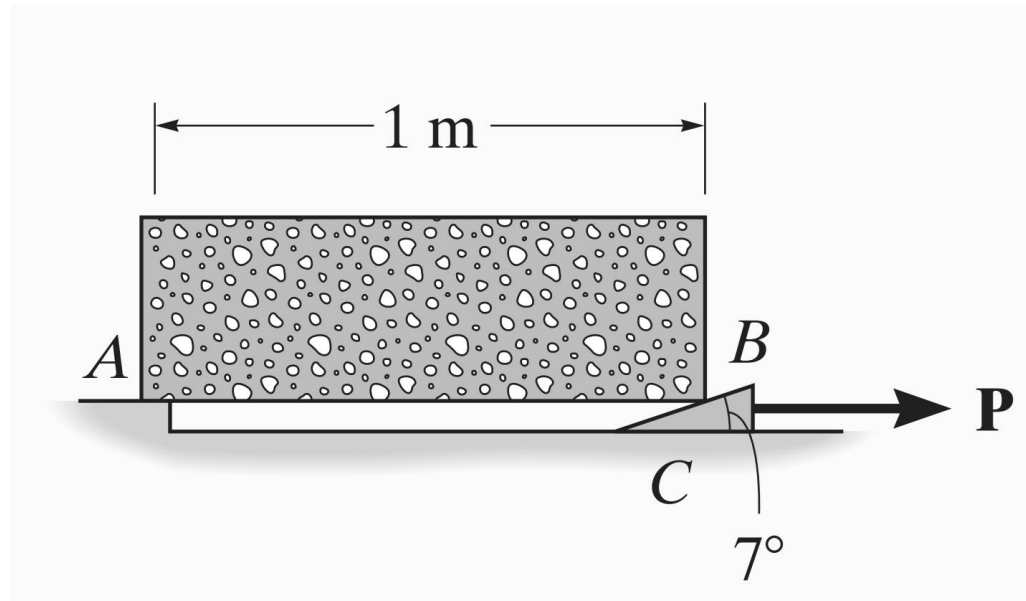
$$P \approx \frac{Wa}{r}$$

The distance a is termed the ***coefficient of rolling resistance***, which has the dimension of length.



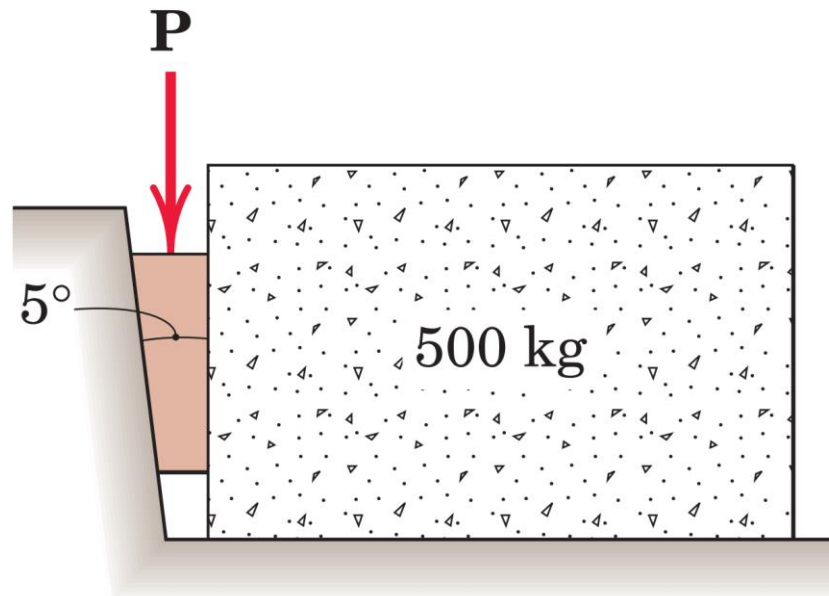
Practice at Home

1. The uniform stone has a mass of 500 kg and is held in the horizontal position using a wedge at B . If the coefficient of static friction is $\mu_s = 0.3$ at the surfaces of contact, determine the minimum force P needed to remove the wedge. Assume that the stone does not slip at A .



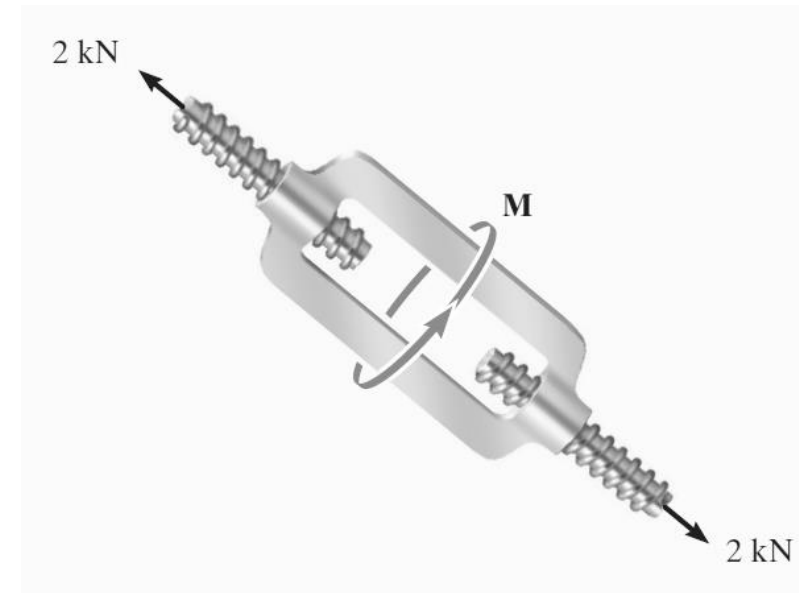
Practice at Home

2. The horizontal position of the 500-kg rectangular block of concrete is adjusted by the 5° wedge under the action of the force $\mathbf{P}$. If the coefficient of static friction for both wedge surfaces is 0.30 and if the coefficient of static friction between the block and the horizontal surface is 0.60, determine the least force P required to move the block.



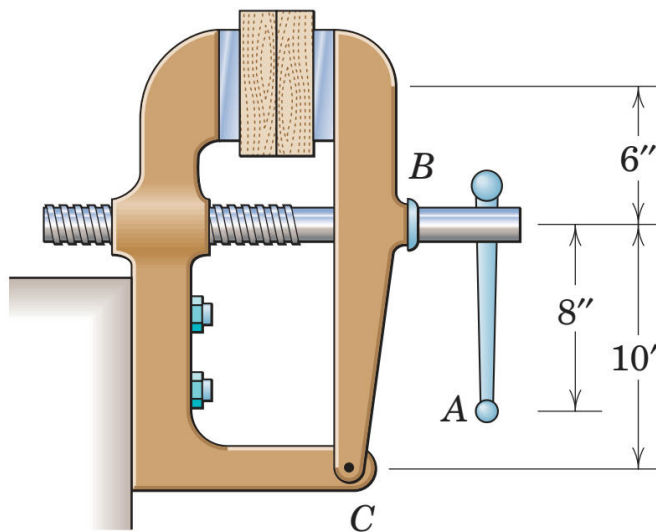
Practice at Home

3. The turnbuckle shown below has a square thread with a mean radius of 5 mm and a lead of 2 mm. If the coefficient of static friction between the screw and the turnbuckle is $\mu_s = 0.25$, determine the moment **M** that must be applied to draw the end screws closer together.



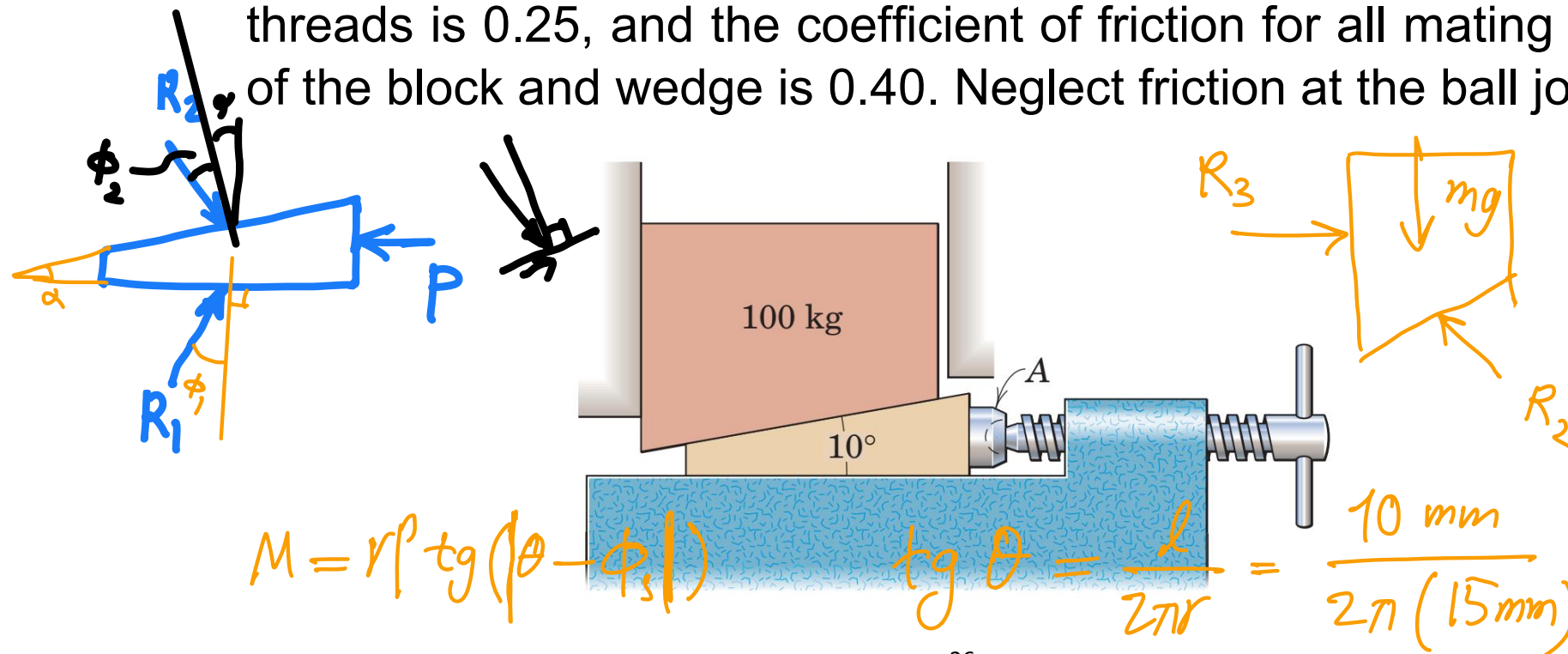
Practice at Home

4. The single-threaded screw of the vise has a mean diameter of 1 in. and has 5 square threads per inch. The coefficient of static friction in the threads is 0.20. A 60-lb pull applied normal to the handle at A produces a clamping force of 1000 lb between the jaws of the vise. (a) Determine the frictional moment M_B , developed at B , due to the thrust of the screw against the body of the jaw. (b) Determine the force Q applied normal to the handle at A required to loosen the vise.



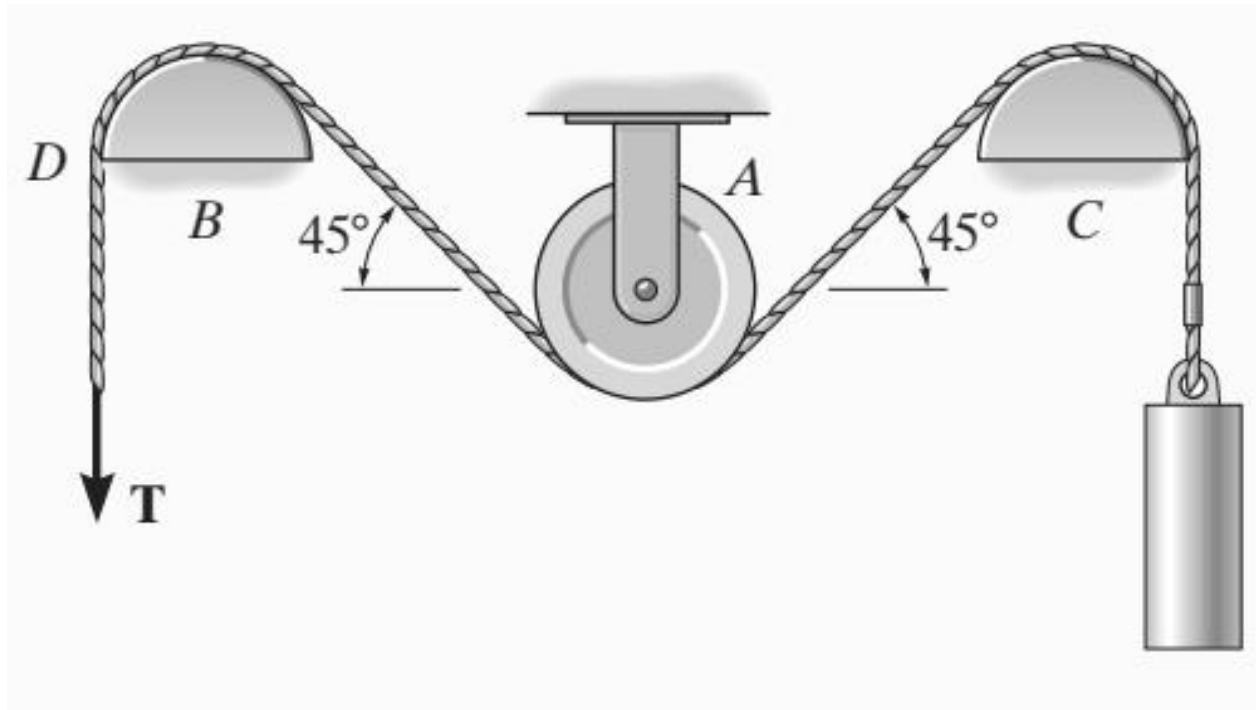
Practice at Home

5. The vertical position of the 100-kg block is adjusted by the screw-activated wedge. Calculate the moment M which must be applied to the handle of the screw to raise the block. The single-thread screw has square threads with a mean diameter of 30 mm and advances 10 mm for each complete turn. The coefficient of friction for the screw threads is 0.25, and the coefficient of friction for all mating surfaces of the block and wedge is 0.40. Neglect friction at the ball joint A.



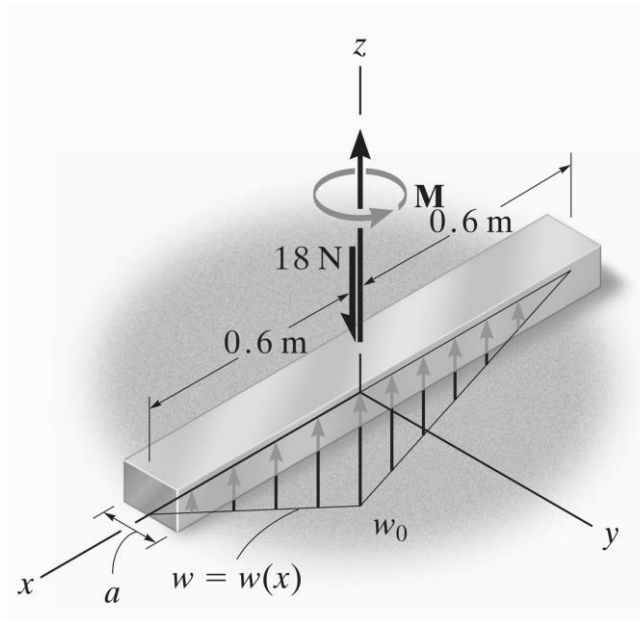
Practice at Home

6. The maximum tension that can be developed in the cord shown in below is 500 N. If the pulley at A is free to rotate and the coefficient of static friction at the fixed drums B and C is $\mu_s = 0.25$, determine the largest mass of the cylinder that can be lifted by the cord.



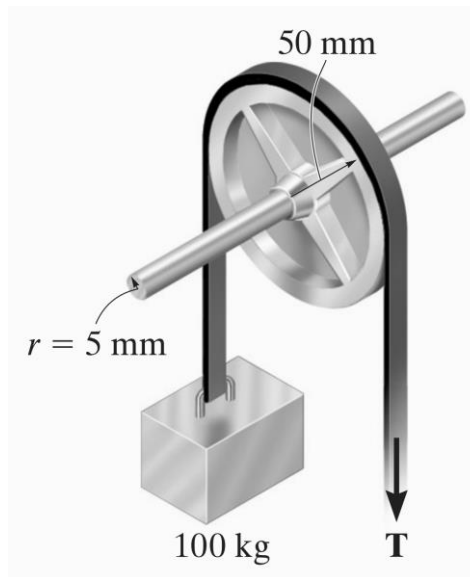
Practice at Home

7. The uniform bar shown below has a weight of 18 N. If it is assumed that the normal pressure acting at the contacting surface varies linearly along the length of the bar as shown, determine the couple moment $\mathbf{M}$ required to rotate the bar. Assume that the bar's width is negligible in comparison to its length. The coefficient of static friction is equal to $\mu_s = 0.3$.



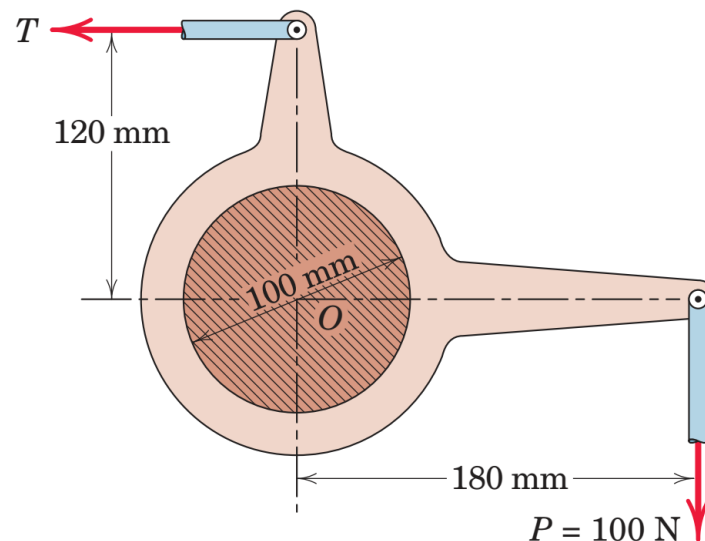
Practice at Home

8. The 100-mm-diameter pulley shown below fits loosely on a 10-mm-diameter shaft for which the coefficient of static friction is $\mu_s = 0.4$. Determine the minimum tension T in the belt needed to (a) raise the 100-kg block and (b) lower the block. Assume that no slipping occurs between the belt and pulley and neglect the weight of the pulley.



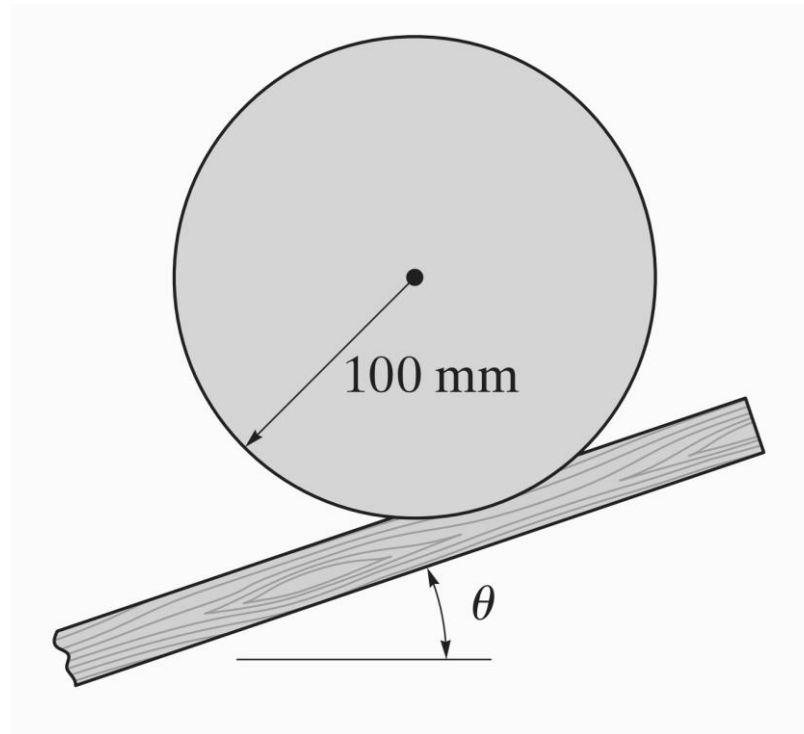
Practice at Home

9. The bell crank fits over a 100-mm-diameter shaft which is fixed and cannot rotate. The horizontal force T is applied to maintain equilibrium of the crank under the action of the vertical force $P = 100$ N. Determine the maximum and minimum values which T may have without causing the crank to rotate in either direction. The coefficient of static friction between the shaft and the bearing surface of the crank is 0.20.



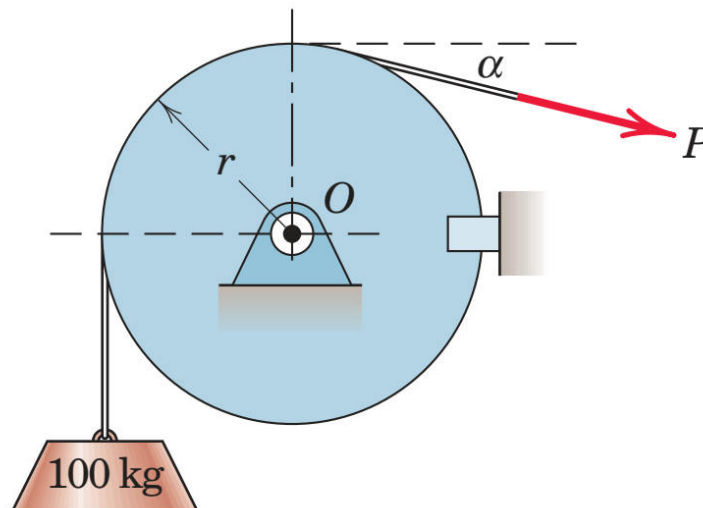
Practice at Home

10. A 10-kg steel wheel shown below has a radius of 100 mm and rests on an inclined plane made of soft wood. If θ is increased so that the wheel begins to roll down the incline with constant velocity when $\theta = 1.2^\circ$, determine the coefficient of rolling resistance.



Practice at Home

11. A flexible cable which supports the 100-kg load is passed over a fixed circular drum and subjected to a force P to maintain equilibrium. The coefficient of static friction between the cable and the fixed drum is 0.30. (a) For $\alpha = 0$, determine the maximum and minimum values which P may have in order not to raise or lower the load. (b) For $P = 500$ N, determine the minimum value which the angle α may have before the load begins to slip.



Practice at Home

12. Determine the range of mass m over which the system is in static equilibrium. The coefficient of static friction between the cord and the upper curved surface is 0.20, while that between the block and the incline is 0.40. Neglect friction at the pivot O.

